

1. Dataset Overview

Total Rows: 300

Total Columns: 18

Domain: ASD Screening

Purpose: Early detection of Autism Spectrum Disorder (ASD) and severity analysis using behavioral and demographic data

2. Column Description

Column Name	Type	Description
Case_ID	Categorical	Unique identifier for each subject
A1–A10	Binary (0/1)	Behavioral questionnaire responses
Age	Numeric	Age of the child (in years)
Gender	Categorical	Male (M) / Female (F)
Ethnicity	Categorical	Ethnic background
Family_History	Categorical	Family history of ASD (Yes/No)
AQ_Score	Numeric	Total AQ score (sum of A1–A10)
Class	Categorical	ASD detection result (ASD / No ASD)
Severity	Categorical	ASD severity level (Mild / Moderate / Severe)

3. Types of Data in the Dataset

3.1 Behavioral Data

A1–A10: Indicators of social behavior, communication, and repetitive patterns.
Classification Class ASD vs Non-ASD detection

Classification Severity Mild, Moderate, Severe classification

Regression

(Optional)AQ_Score Predict screening score

Task TypeTarget

VariableDescription

Classification: Class ASD vs Non-ASD detection Classification:

Severity Mild, Moderate, Severe classification

Regression :AQ_Scor

(Optional) ePredict screening score

Demographic Data

Age: Age of the subject Gender: Gender

of the subject Ethnicity: Background

information Family_History: Genetic risk

indicator 3.3 Clinical / Derived Data

AQ_Score : Computed screening score

Class: ASD detection result Severity: Level

of ASD condition 4. Target Variables

5. Severity Levels

AQ Score Range Severity

0 – 3 Mild

4 – 7 Moderate

8 – 10 Severe

{Reference Dataset Links}

<https://www.kaggle.com/datasets/faizunnabi/autism-screening>

<https://www.kaggle.com/datasets/hbunyamin/autistic-spectrum-disorder>

<https://github.com/topics/autism-spectrum-disorder>

https://github.com/Nandini-singh05/Autism_Detection_using_Machine_Learning/blob/main/autism_data.csv

We used these datasets and improved them by adding behavioral, communication, and temporal features for severity prediction and longitudinal analysis.